eventually merge into the host's tissues and become unrecognizable as a parasite at all. Maybe, as I suggested on page 237, our cells have come far across this evolutionary spectrum: we are all relics of ancient parasitic mergers.

Look at what can happen when parasite genes and host genes do share a common exit. Wood-boring ambrosia beetles (of the species *Xyleborus ferrugineus*) are parasitized by bacteria that not only live in their host's body but also use the host's eggs as their transport into a new host. The genes of such parasites therefore stand to gain from almost exactly the same future circumstances as the genes of their host. The two sets of genes can be expected to 'pull together' for just the same reasons as all the genes of one individual organism normally pull together. It is irrelevant that some of them happen to be 'beetle genes', while others happen to be 'bacterial genes'. Both sets of genes are 'interested' in beetle survival and the propagation of beetle eggs, because both 'see' beetle eggs as their passport to the future. So the bacterial genes share a common destiny with their host's genes, and in my interpretation we should expect the bacteria to cooperate with their beetles in all aspects of life.

It turns out that 'cooperate' is putting it mildly. The service they perform for the beetles could hardly be more intimate. These beetles happen to be haplodiploid, like bees and ants (see Chapter 10). If an egg is fertilized by a male, it always develops into a female. An unfertilized egg develops into a male. Males, in other words, have no father. The eggs that give rise to them develop spontaneously, without being penetrated by a sperm. But, unlike the eggs of bees and ants, ambrosia beetle eggs do need to be penetrated by *something*. This is where the bacteria come in. They prick the unfertilized eggs into action, provoking them to develop into male beetles. These bacteria are, of course, just the kind of parasites that, I argued, should cease to be parasitic and become mutualistic, precisely because they are transmitted in the eggs of the host, together with the host's 'own' genes. Ultimately, their 'own' bodies are likely to disappear, merging into the 'host' body completely.

A revealing spectrum can still be found today among species of hydra —small, sedentary, tentacled animals, like freshwater sea anemones. Their tissues tend to be parasitized by algae. (The 'g' should be pronounced hard. For unknown reasons some biologists, not least in America, have recently taken to saying Algy as in Algernon, not only for